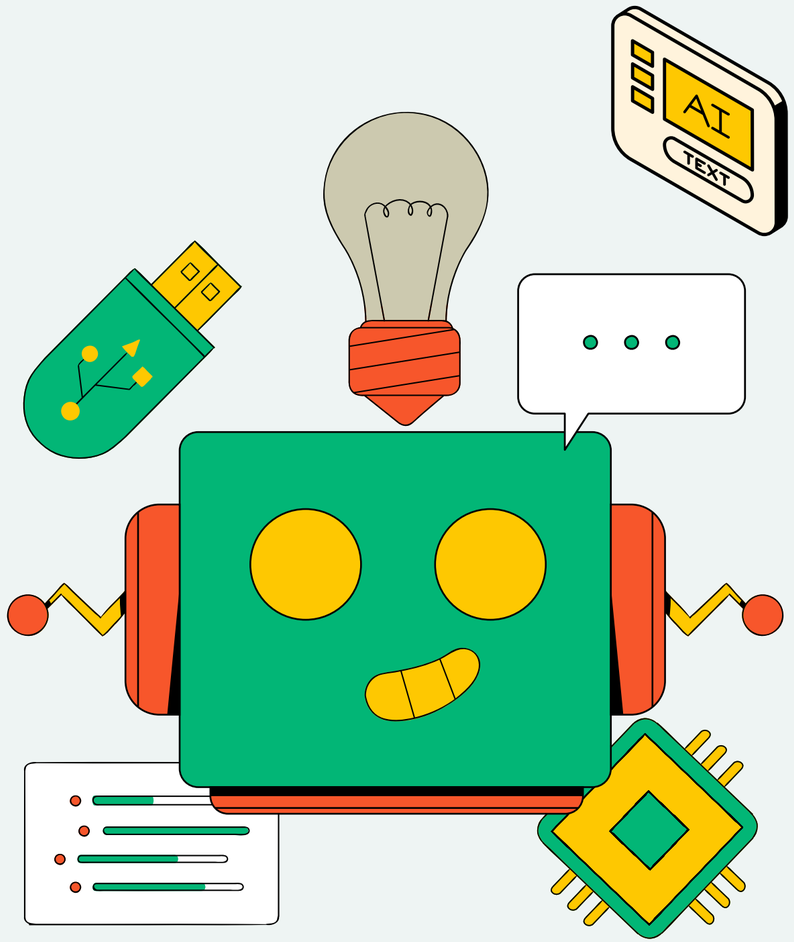


TRAFFIC CONGESTION PREDICTION IN URBAN AREAS USING MACHINE LEARNING MODELS PRESENTATION

PRESENTED BY:

GROUP 9

PRESENTATION OUTLINE

- Introduction
- Problem Statements & Objectives
- Related Works
- Datasets & Preprocessing
- Methodology & Model Selection
- Experimental Setup & Training
- Results, Evaluation & Discussion
- Solution & Business Proposal
- Conclusion

INTRODUCTION

- Urban traffic congestion is a growing issue in modern cities
- Traditional traffic control methods lack real-time adaptability
- Rise of smart city technologies enables large-scale traffic data collection
- Machine learning offers powerful tools for congestion prediction

Motivation: Improve urban mobility using accurate, scalable, and interpretable ML models

PROBLEM STATEMENTS & OBJECTIVES

Problem Statements	Objectives
Inefficiency of Traditional Traffic Management Systems	To develop and evaluate machine learning models for urban traffic congestion prediction
Underutilization of Heterogeneous and High-Volume Traffic Data	To preprocess and integrate diverse traffic datasets for improved model readiness.
Limited Generalizability of Machine Learning Models Across Different Urban Contexts	To perform cross-dataset performance analysis to assess model generalization.
Insufficient Emphasis on Feature Interpretability and Model Transparency	To analyze the importance of predictive features to enhance model interpretability.

RELATED WORKS

Study	Dataset Source	Models Used	Performances (Accuracy)	Limitations
Zafar & Ul Haq (2020)	Google Maps API + Weather (Islamabad)	RF, XGBoost, SVM	RF: 92% XGBoost: 91% SVM: 39%	Coarse weather data; not generalizable
Arabiat et al. (2024)	Intersection Sensors (Amman)	SVM, RF, DT, LR, KNN, MLP	SVM: 99.1–99.7%, RF: ~98%, DT: ~96%, KNN: ~96%, MLP: 94.5–96.3%, LR: ~95%	Single-location; overfitting risk
Vineetha et al. (2025)	GPS + Sensors + Weather	RF	87%	Hardware-dependent; complex integration
Hossain et al. (2022)	Kaggle GPS & PVD	LSTM, RF, SVM, KNN, XGBoost	LSTM: 98%, KNN: 96%, SVM & XGBoost: 94%	Dataset quality issues
Suganyadevi et al. (2023)	VANET datasets	RF, SVM, KNN	RF: 97%	Privacy & connectivity issues
Türkan & Ulu (2025)	Istanbul traffic + patrol data	RF, DT, KNN	Not specified	Limited urban dataset adaptability
Liu & Wu (2017)	Shanghai traffic + weather	RF	RF: 87.5%	Small sample; single city
Mystakidis & Tjortjis (2020)	Open APIs + GIS (Greece)	DT, LR	DT: 73%	Limited data coverage; scalability issues
Tamir et al. (2020)	Highway sensor network	DT, LR, NN	DT: 97.65% (Clementine), 97% (Python)	Sensor-specific; limited context transfer
Kanchi et al. (2022)	U.S. Accident Data	RF, DT, KNN, NN	RF: 90%	Geographic generalizability challenges

DATASETS

Dataset 1: US Traffic Congestions (2016–2022)

Dataset 2: Smart Mobility Traffic Dataset

Dataset 3: Traffic Prediction Dataset

DATASETS & PREPROCESSING

DATASET 1:

- Source: Kaggle
- Samples: ~2 million events (sampled from ~33M)
- Features: Severity, GPS, time, weather, delay
- Preprocessing: Feature reduction, time parsing, missing value imputation, scaling, class balancing

Features	Description
ID	Unique identifier of the congestion event
Severity	Event severity level (0 = least, 4 = most severe) – target variable
Start_Lat, Start_Lng	Approximate GPS coordinates of event start
StartTime, EndTime	Start and end time of the congestion event (local timezone)
Distance(mi)	Road length affected by the congestion event
DelayFromTypicalTraffic(mins)	Delay compared to typical traffic flow
DelayFromFreeFlowSpeed(mins)	Delay compared to free-flow (ideal) speed
Congestion_Speed	Ranked traffic speed level (categorical)
Description	Human-provided text description of the event
Street, City, County	Geographical location fields (from reverse geocoding)
State, Country, ZipCode	Administrative location fields
LocalTimeZone	Time zone of the event
WeatherStation_AirportCode	Code of nearest weather station or airport
WeatherTimeStamp	Timestamp of recorded weather conditions
Temperature(F)	Temperature in Fahrenheit
WindChill(F)	Wind chill temperature in Fahrenheit
Humidity(%)	Humidity percentage
Pressure(in)	Atmospheric pressure in inches
Visibility(mi)	Visibility distance in miles
WindDir	Wind direction
WindSpeed(mph)	Wind speed in miles per hour
Precipitation(in)	Precipitation amount in inches
Weather_Event	Type of weather event (e.g., Rain, Fog, Storm)
Weather_Conditions	Overall weather condition (e.g., Clear, Snowy, Cloudy)

PREPROCESSING

Dataset 1: US Traffic Congestions (2016–2022)

1. Initial Cleaning: Dropped redundant and mostly-null columns (e.g., Description, Weather_Event)
2. Datetime Engineering: Extracted Hour, DayOfWeek, Month from StartTime
3. Missing Values: Median imputation for numerics; 'Unknown' for categoricals
4. Rare Class Grouping: Weather conditions <1000 grouped as 'Other'
5. Encoding: Ordinal for Congestion_Speed; one-hot for WindDir, Weather_Conditions
6. Scaling: StandardScaler (weather, GPS), MinMaxScaler (delays)
7. Imbalance: Removed Severity = 4 due to extreme underrepresentation
8. Train-Test Split: 80:20 ratio

DATASETS & PREPROCESSING

DATASET 2:

- Source: Kaggle
- Samples: ~200,000 records (approx.)
- Features: Vehicle count, speed, occupancy, sentiment, weather, energy
- Preprocessing: Datetime extraction, one-hot encoding, scaling, ordinal target encoding

Features	Description
Timestamp	Date and time of the recorded observation
Latitude, Longitude	GPS coordinates of the event or traffic point
Vehicle_Count	Number of vehicles at the given location and time
Traffic_Speed_kmh	Average traffic speed in kilometers per hour
Road_Occupancy_%	Road space usage percentage
Traffic_Light_State	Current state of the traffic light (e.g., Red, Green)
Weather_Condition	Categorical description of weather (e.g., Clear, Rainy)
Accident_Report	Binary indicator of an accident occurring at the same time/location
Sentiment_Score	Public sentiment score based on social media data
Ride_Sharing_Demand	Level of demand for ride-sharing services
Parking_Availability	Number or percentage of parking spaces available
Emission_Levels_g_km	CO_2 emissions in grams per kilometer
Energy_Consumption_L_h	Energy consumption in liters per hour
Traffic_Condition	Categorized congestion level: Low, Medium, or High – target

PREPROCESSING

Dataset 2: Smart Mobility Traffic Dataset

1. Datetime Parsing & Features Extraction: Parsed Timestamp; extracted Hour, DayOfWeek, Month
2. Categorical Encoding: One-hot for Traffic_Light_State, Weather_Condition
3. Target Encoding: Ordinal mapping for Traffic_Condition
4. Feature Scaling: StandardScaler on numerical variables
5. Train-Test Split: 80:20 ratio

DATASETS & PREPROCESSING

DATASET 3:

- Source: Kaggle
- Samples: ~10,000 records (15-min intervals)
- Features: Vehicle counts by type, total, timestamp
- Preprocessing: Synthetic datetime, temporal features, scaling, target label encoding

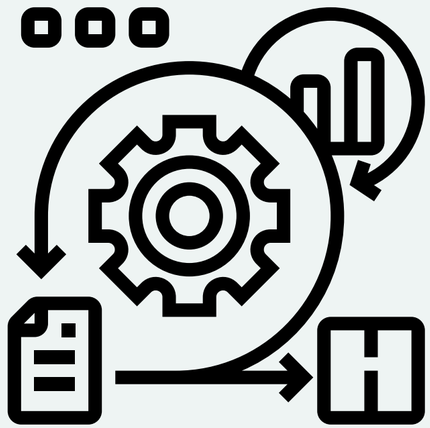
Features	Description
Timestamp	The specific time of the observation (in HH:MM:SS format).
Date	The day of the month (1–31); year and month were not explicitly provided.
Day of the week	Categorical day name (e.g., Monday, Tuesday).
CarCount	Number of cars detected.
BikeCount	Number of bikes detected.
BusCount	Number of buses detected.
TruckCount	Number of trucks detected.
Total	Sum of all vehicle types detected in the interval.
Traffic Situation	Target label representing traffic conditions, classified into one of four congestion levels: Heavy, High, Normal, or Low.

PREPROCESSING

Dataset 3: Traffic Prediction Dataset

1. Temporal Features Extraction: Combined day + time as datetime; extracted Hour, Minute, DayOfWeek, Month
2. Target Encoding: Ordinal encoding for Traffic Situation (low to heavy)
3. Scaling: StandardScaler on all vehicle count features
4. Train–Test Split: 80:20 ratio

Methodology and Model Selection



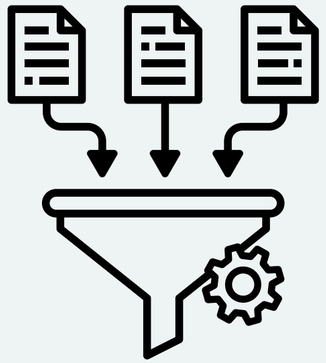
Methodology Overview:

- 3 real-world traffic datasets used
- Goal: Predict traffic congestion levels/severity
- Applied 5 machine learning models
- Compared performance using accuracy & F1-score

Step-by-Step Process:

- Collected and understood datasets
- Preprocessed data (cleaning, encoding, scaling)
- Split data (train/test)
- Trained models using GridSearchCV
- Evaluated models with multiple metrics

Methodology and Model Selection



Data Preprocessing:

- Handled missing values
- Created new features from time/date
- Encoded categorical values
- Scaled numeric data

Machine Learning Models:

- Random Forest
- Logistic Regression
- XGBoost
- Decision Tree
- LinearSVC

Methodology and Model Selection

Why These Models?

- Popular in traffic prediction research
- Good accuracy and generalization
- Some are fast & simple (LogReg, Decision Tree)
- Others handle complex patterns (XGBoost, RF)
- Based on assignment literature review

Model Training:

- 80% training, 20% testing split
- GridSearchCV used for tuning
- Cross-validation applied

Evaluation Metrics

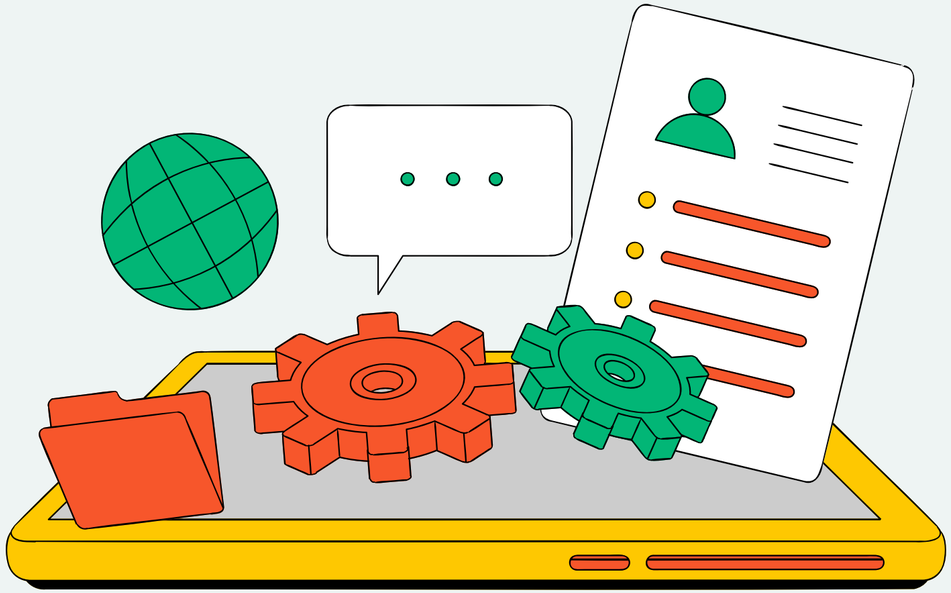
- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

EXPERIMENTAL SETUP AND TRAINING

Data Preprocessing

Model Architecture

Hyperparameter Tuning



Experimental Setup and Training

Data Preprocessing :

- Feature Scaling
  - StandardScaler, MinMaxScaler
- Feature Selection
  - Use Random Forest to obtain importance score
- Train and Test split
  - 80:20 ratio

: Model Architecture

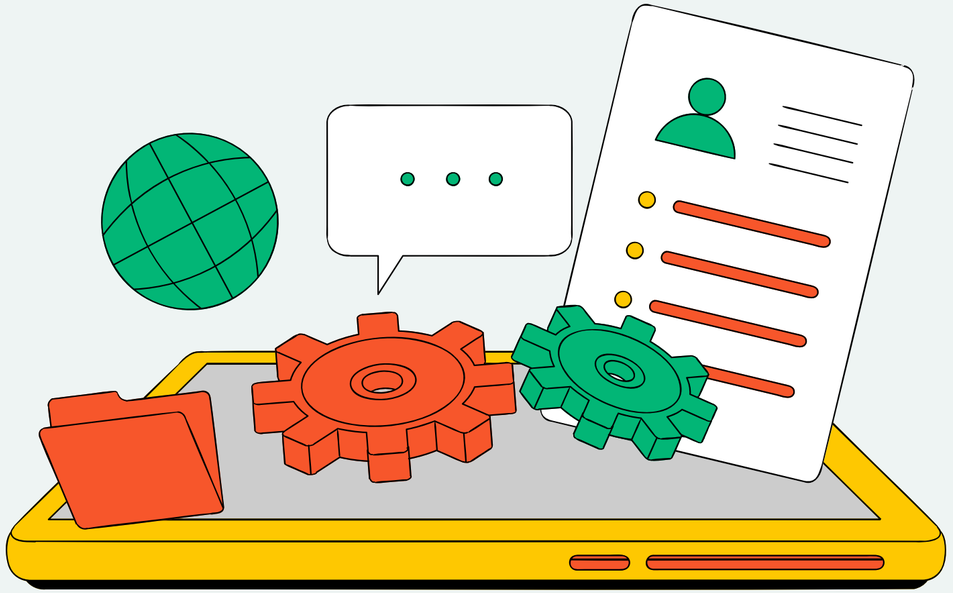
- Random Forest, XGBoost
  - n_estimators=100, max_depth=10 and random_state=42
- Logistic Regression
  - max_iter=1000 and solver=lbfgs
- Decision Tree
  - max_depth=10 and random_state=42
- LinearSVC
  - max_iter=10000

Hyperparameter Tuning

- RandomizedSearchCV
  - US Traffic Congestions (2016–2022) dataset (except SVC)
- GridSearchCV
  - US Traffic Congestions (2016–2022) dataset (SVC)
  - Smart Mobility Traffic
  - Traffic Prediction datasets

EVALUATION, DISCUSSION & CONCLUSION

- Evaluation
- Accuracy Analysis
- Discussion
- Benchmarking
- Limitation and Improvement



ACCURACY ANALYSIS

Models	Accuracy		
	Dataset 1	Dataset 2	Dataset 3
Random Forest	0.72 (Tuned)	1.00 (Tuned)	1
Logistic Regression	0.52	0.77 (Tuned)	0.88 (Tuned)
XGBoost	0.78	1	1
Decision Tree	0.72 (Tuned)	1.00 (Tuned)	1.00 (Tuned)
SVC	0.53 (Tuned)	0.75 (Tuned)	0.85 (Tuned)

BENCHMARKING

Comparison		
Grp 9	**Model**	**Source**
1	Random Forest	1
0.77 (Tuned)	Logistic Regression	0.72
1	XGBoost	1
1	Decision Tree	1
0.75	SVC	0.63

BENCHMARKING

Comparison		
Grp 9	**Model**	**Source**
1	Random Forest	0.99
0.88 (Tuned)	Logistic Regression	0.9
1	XGBoost	1
1	Decision Tree	-
0.85 (Tuned)	SVC	0.91

DISCUSSION

US TRAFFIC CONGESTIONS (2016-2022) DATASET

Linear models: low accuracy (50–53%)
Tree-based models performed better (>70%)
Faced memory issues during tuning → reduced to cv=2, n_jobs=1

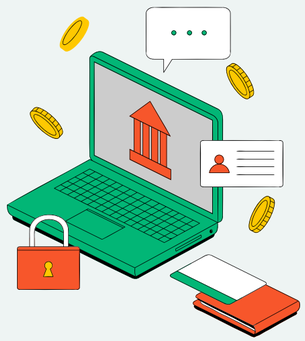
SMART MOBILITY TRAFFIC DATASET

Ensemble models: 100% accuracy
Log-R: 77% accuracy
SVC: 75%
Simple, clean datasets allow strong performance with minimal tuning

TRAFFIC PREDICTION DATASET

Ensemble models: 100% accuracy
Log-R: 88% accuracy
SVC: 85%
Linear models have lower accuracy due to fewer temporal features

LIMITATIONS AND IMPROVEMENTS



LOWER PERFORMANCE ON LARGE DATASET

- US dataset: lower accuracy due to complexity
- Resource constraints limited hyperparameter tuning (cv=2, n_jobs=1)
- **Improvement**: Use scalable tuning methods (Successive Halving, Bayesian Optimization)

BASIC FEATURE ENGINEERING

- Relied mainly on Random Forest importance
- No external data like holidays, events, real-time incidents
- **Improvement**: Richer contextual features (Add rush hour flags, day type, holiday/event indicators, combine weather features)

LACK OF TEMPORAL & SPATIAL MODELING

- In real life, traffic is influenced by patterns (rush hours, events) which can affect nearby roads
- May miss important patterns and make less accurate predictions without these patterns
- **Improvement:** Use deep learning for spatio-temporal learning (LSTM, GNNs, Hybrid)

SOLUTION & BUSINESS PROPOSAL

- Real-World Integration
- Scalability and Performance
- Target Users
- Commercial Value and SWOT Analysis

REAL-WORLD INTEGRATION

Integrate into urban infrastructure, municipal transport system or third-party apps.

Core Functionality:

- Uses machine learning models trained on historical and real-time traffic data.
- Predicts congestion levels: Low, Medium and High
- Forecasts future traffic patterns with high accuracy.

Use Cases:

- Urban Traffic Control Centers
- Smart City Platforms
- Navigation Apps (e.g. Waze and Google Maps)
- Logistics & Delivery Services
- Public Transport Systems

SCALABILITY AND PERFORMANCE

GENERAL FEATURES:

- Modular system – adaptable to both small and large cities
- Works with varied data availability and sources
- Lightweight models (Random Forest, Decision Tree, Logistic Regression)
- Efficient for retraining and deployment

SCALABILITY FEATURES:

- Edge: Local predictions via sensors & traffic lights
- Cloud: Scalable API via Huawei Cloud
- Updates: Auto-retrain with new traffic data

TARGET USERS

Government Agencies
- Traffic control, urban planning, municipal IT

Private Sector
- Logistics, ride-hailing, delivery, smart infra providers

Public Commuters
- Bus riders, navigation app users, public transport users

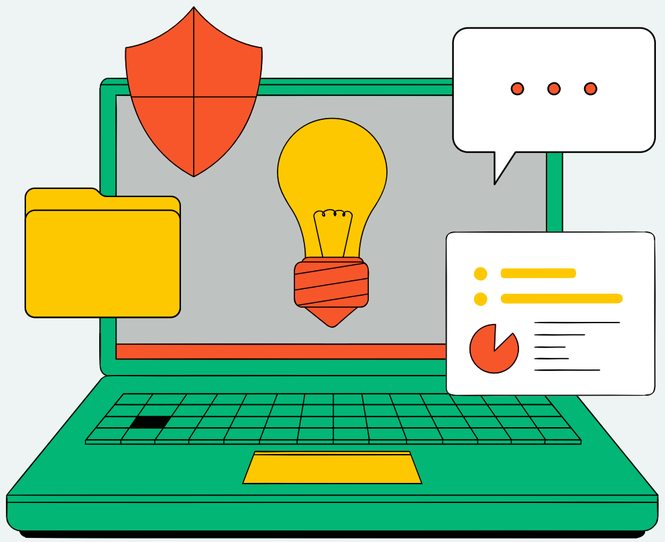
COMMERCIAL VALUE & SWOT ANALYSIS

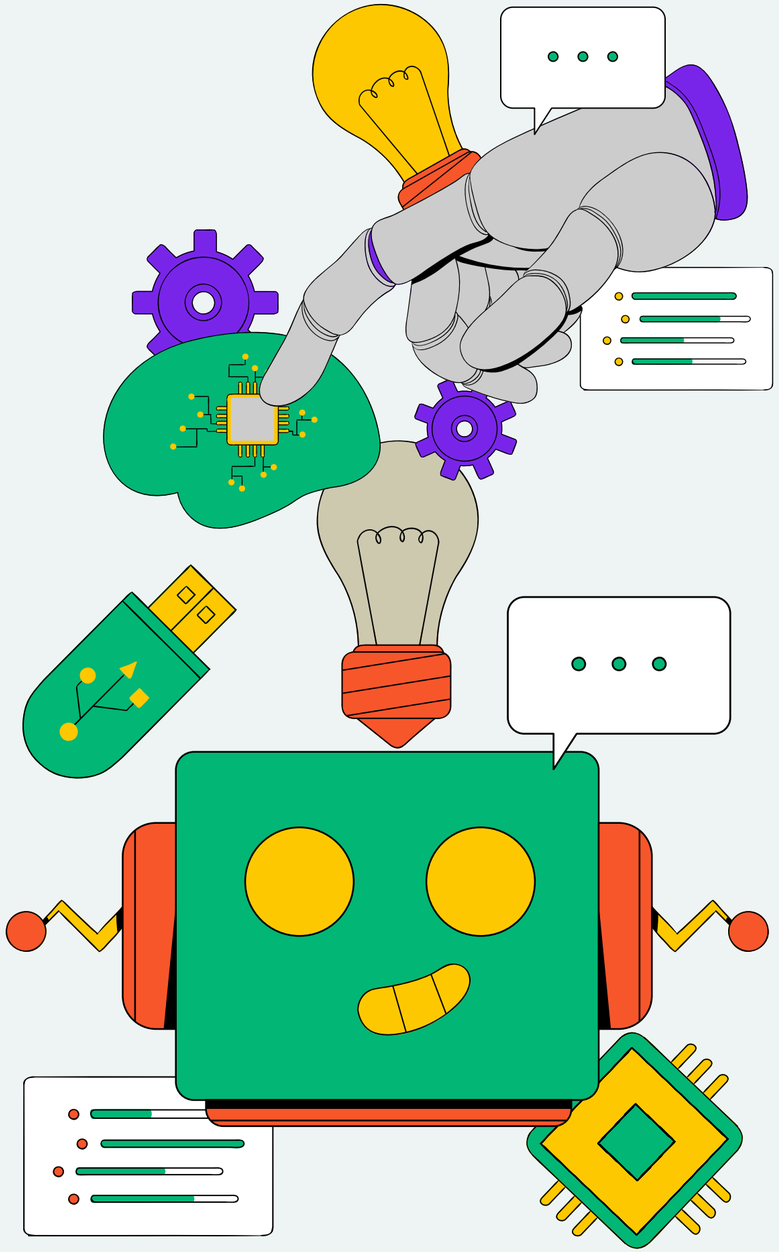
- **strengths enable wide deployment and integration in smart city environments**
- **Weaknesses must be planned for with robust data infrastructure**

🟢 **Strengths**	🔴 **Weaknesses**
✅ Proven ML models	⚠️ Needs high-quality, real-time traffic data
☁️ Huawei cloud & edge compatibility	🛠️ Infrastructure dependency (sensors,
🔌 Easy API integration	

🟡 **Opportunities**	⚫ **Threats**
🌆 Growth in smart city tech investment	🏢 Competition from major tech firms
🚗 AI in intelligent transportation	🔐 Privacy & data regulation challenges
🏛️ Government digital transformation	

CONCLUSION



Limitations	Future Work
Relies on large, clean, labeled datasets for high accuracy	Integrate **deep learning models** (e.g., LSTM, GNN, Transformers) to improve pattern recognition
Performance drops in areas with **noisy or sparse data**	Use **multi-modal data sources** (GPS, LiDAR, CCTV, weather feeds) for better prediction accuracy
Struggles with **unpredictable real-time events** like accidents or roadblocks	Develop **event-aware models** that respond to live incident reports
Limited scalability in regions with poor infrastructure (IoT, network access)	Expand **edge deployment** using lightweight AI models on traffic controllers and local servers
Classical ML may miss complex spatiotemporal relationships	Improve **model interpretability** with tools like SHAP or LIME for better decision support

THANK YOU